

ARIHANT JAIN

Senior Backend Engineer | Distributed Systems

New Delhi, India | +91 8094221469 | arihantjain416@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

Senior Software Engineer

Apr 2024 – Present

Perfios

New Delhi, IN (Remote)

- Delivered **\$700K+ in cumulative cost savings since Apr 2024** by reducing vendor dependence across KYC and utility workflows through caching, source orchestration, and workflow redesign.
- Owned active-passive disaster recovery across Mumbai and Hyderabad for TKYC, supporting **5M+ daily requests at over 99.9% measured availability**; expanded the suite internationally with **20+ KYC APIs** across Kenya, UAE, and Indonesia.
- Migrated KYC and utility services from Python 3.9 to 3.12, containerizing **245 of 260 APIs** for cloud and on-premises deployment with **zero production rollbacks**.
- Built GitLab CI/CD pipelines with SAST for multi-region deployments across multiple Lambda functions, replacing a dedicated EC2 deployment VM.
- Automated browser-based PDF parsing and generation with Playwright, increasing success rate from **60% to 98%** and reducing average latency from **40s to 6s**.
- Piloted the first migration to a shared data-ingestion framework with improved observability, then scaled the reviewed pattern across **20+ services** using AI-assisted workflows while preserving critical fallback behavior.
- Conducted **100+ code reviews** and mentored junior and support engineers on debugging and incident triage.

Software Engineer I

Oct 2022 – Apr 2024

Karza Technologies (merged into Perfios)

Mumbai, IN (Remote)

- Re-architected a Lambda-based source as a Redis-backed Celery worker on AWS Fargate; serialized vendor-session handling eliminated login churn and tripled throughput, while caching and rate limits enabled vendor retirement, saving **\$230K+ over two years**.
- Developed **30+ APIs** within six months and maintained **100+ production APIs**, supporting integrations across **300+ public registries** and services handling **1M+ daily requests**.
- Re-engineered legacy KYC APIs, reducing **P99 latency from 8s to 5s** and strengthening reliability for client-facing verification flows.
- Reduced database writes from **9–10 to 3–4 per request** on APIs serving up to **500K daily requests**; built vendor-billing and scheduled data-analysis tools to improve billing accuracy.

Software Engineer

Mar 2021 – Sep 2022

InTimeTec

Jaipur, IN (Onsite)

- Drove a **0-to-1 modernization** of a legacy .NET desktop application into an Angular web platform across **9 modules**, tripling operational throughput.
- Led end-to-end delivery while coordinating a **4-engineer pod**; reduced deployment cycles by **44%** and increased test coverage by **54%** through CI/CD and automated testing.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL

Frameworks & APIs: FastAPI, Node.js, REST APIs, Pydantic

Data & Messaging: Redis, MongoDB, Celery, Kafka, RabbitMQ, Amazon SQS

Cloud & Delivery: AWS Lambda, Fargate, ECS, S3, EC2, CloudFormation, Docker, Linux, GitLab CI/CD

Testing, Security & Observability: Pytest, Playwright, Selenium, AWS CloudWatch, VAPT workflows

PROJECTS

Distributed Quota-Metering Engine | *FastAPI, Redis Lua, MongoDB, Docker Compose*

[GitHub](#)

- Engineered a per-organization per-feature quota engine using Redis Lua for atomic idempotency checks, validation, deduction, and denial recording.
- Implemented all-or-nothing batch requests, idempotent refunds, and monthly UTC resets without scheduled jobs.
- Load-tested **10,000 simulated requests** with **zero over-serving**; same-key retries consumed quota once.

EDUCATION

Swami Keshvanand Institute of Technology

Aug 2017 – Jun 2021

B.Tech, Information Technology

Jaipur, IN